TEAM NEURAFORGE · SIH26024

What changed, and which slide it lands on

Every claim below is measured on the running system, not estimated. Tags say whether a slide gains a claim, keeps a claim with a better number, or has to be corrected because the old wording is no longer true.

5 September 2026
Gevra OC · Korba
APK **2.07 GB** · 4 models
Chain **intact**, 4 records
18 monitoring points

Figures the deck can quote

All measured today on the Galaxy S24+ and the local stack. Where a number replaces one already on a slide, the entry below says so.

4 / 4 GROUNDED ANSWERS correct owner, date and clause	**1,078** TRAINING EXAMPLES from the project's own corpus	**95.9%** TOKEN ACCURACY 6 epochs, loss 0.352
238 MB TUNED MODEL Q4_0, runs on the phone's GPU	**410.6 ha** OUTSIDE THE LEASE PostGIS ST_Difference	**18** MONITORING POINTS named by regulation, not invented
87,302 SENSOR READINGS 5 live streams, 51 alerts		

The comparison slide writes itself

Same question, three answerers, measured within an hour of each other. This is the strongest new argument for running the model on the handset — and it is a measurement, not a claim.

ANSWERER	WHERE IT RUNS	TIME TO ANSWER	OUTCOME
ANUPALAN 270M	On the phone, airplane mode	`~20 s`	**4 duties, owners and dates correct, clauses cited**
Gemini 2.5-flash-lite	Cloud, paid tier	`0.84 s`	Full answer, but the ledger never leaves the device
NVIDIA `kimi-k3`	Cloud, free tier	`115–144 s`	Two minutes to return the word “OK”
NVIDIA `deepseek-v4-flash`	Cloud, free tier	`160 s, then timeout`	Queue depth, not model size — thinking off changed nothing

The line for the slide: **a queued free tier was six times slower than the phone in the officer's hand**, and the phone needed no network at all.

Slides that need editing

NEW CLAIM

Live demo · headline

The alert loop now closes, and names who was told

Reading → threshold arithmetic → alert → a manager issues an instruction → it appears on every handset at that mine → an officer acknowledges → the dashboard records the name and the time. Demonstrated end to end: a methane breach sent from the phone produced *WITHDRAW ALL PERSONS* on the handset within twelve seconds, acknowledged, and logged. **An alert nobody acted on and an alert somebody stood down look identical on a sensor chart — only this record tells them apart**, which is the distinction an inquiry turns on.

NEW CLAIM

Sensors

Readings now carry the place regulation names

CMR 2017 sets the methane limit in the general body of return air *of a district*; CPCB conditions specify dust at the lease boundary; a blast vibration limit is measured at the nearest house, not at the blast. Eighteen statutory monitoring points are catalogued and every reading and alert carries one, so the alert text reads **"methane 1.94% exceeds the 1.25% trigger at Return airway — District 1"**. Breaches are now tracked per place, so a district-level event can no longer be hidden by one at the main return.

NEW CLAIM

Roadmap / Hardware

No transmitters yet — and the demo says so

The handset carries a sensor console that posts readings by hand at normal, elevated or breach level for any of eight streams. It is labelled as manual entry on the screen itself, and it posts to **exactly the endpoint a plant gateway would use** — nothing is special-cased server-side, so the threshold arithmetic, the clause and the alert are the production path. Wiring a real transmitter replaces that one screen and nothing else. Say this plainly to judges; it is a stronger claim than a canned demo, and it is only true because the server cannot tell the difference.

CORRECTION

Architecture

The runtime is llama.cpp, not LiteRT-LM

Any slide naming **Gemma 4 E4B via LiteRT-LM** is out of date. The app now runs `llama.rn` (llama.cpp) with GGUF weights. This was not cosmetic: LiteRT's OpenCL path produced garbage on this handset's Mali GPU, and llama.cpp's does not. Three models ship on-device — Qwen3 1.7B for the ledger, ANUPALAN 270M for speed, LFM2.5-VL 450M for photographs.

CORRECTION

Evidence integrity

The hash-chain claim is only now actually true

The chain covered `photo_sha256`, coordinates, timestamp and obligation — **but not the officer's written observation**. Editing a finding in the database passed verification cleanly. Measured, not theorised. The canonical string now includes the observation and pins the timestamp to UTC, and a tampered row is caught and named. If a judge asks "can I change what the officer wrote?", the answer is now no.

NEW CLAIM

AI / Model

We trained our own model on our own corpus

Gemma 3 270M fine-tuned into **ANUPALAN 270M** on 1,078 examples built from this project's clause corpus, ledger rows and sensor windows. It answers grounded ledger questions 4/4 with correct owners, dates and citations. Bare recall stays weak, so facts are retrieved and placed in the prompt — the project's own rule applied one level up: **the model does language, deterministic code supplies the facts.**

NEW CLAIM

Evidence / Demo flow

Evidence is now geographically provable

A capture carries its GPS fix; PostGIS `ST_Contains` returns the verdict; the dashboard plots it. Three captures sit inside the Gevra lease in green, one outside in red. **The colour comes from the geometry, not from what the officer typed.**

NEW CLAIM

Dashboard

The register is now managed from the dashboard

Duties can be created, edited, **reassigned** and deleted from the browser, and the handset picks the change up on its next fetch — verified end to end by reassigning a duty and watching it change on the phone. Plus a new Sensors page carrying all five live streams with their thresholds, z-scores and governing clauses.

NEW CLAIM

Safety / Governance

The system refuses things, and says why

Deleting a duty that still has evidence returns **409**: *“Evidence outlives the duty it was captured for.”* A regulator editing the register returns **403** — they inspect, they do not edit. Both are demonstrable in ten seconds, and refusals are more convincing to a compliance audience than features.

BETTER NUMBER

Cost / Offline

Offline is now the fast path, not the compromise

The deck previously argued on-device on grounds of cost and connectivity. Add performance: the phone answered a real ledger question in about 20 seconds while a free cloud tier took two minutes to say “OK”. An optional multi-provider cloud fallback exists, is off by default, and **cannot receive ledger rows by construction** — its function signature has nowhere to put them.

CORRECTION

Fine-tuning

“Granite 4.0 350M” was a leftover label

The fine-tune page defaulted to a base model this project has never trained — a stale string in a free-text box. It now offers only bases actually used. If a screenshot of that page is in the deck, **retake it.**

Two things to be careful about

Do not put the State Emblem or Ministry of Coal mark in the deck

Its use is restricted by the **State Emblem of India (Prohibition of Improper Use) Act, 2005**. A student prototype carrying it asserts government authority the project does not have — at a government-run competition, in front of judges who will recognise it immediately. The risk is not a takedown; it is a compliance tool caught misrepresenting its own standing, which undercuts the whole pitch.

Three images in the build have unconfirmed provenance

The login and Overview photographs are Pexels and Pixabay, free for commercial use. The Duties, Capture and Sync photographs carry no watermark but no known licence either. Fine for a demo; swap them for Pexels equivalents before anything public. Images that still carry a Chegg logo, a Shutterstock ID or a Freepik watermark were left out.

Still open

- **Deck screenshots are stale.** The dashboard gained a navy header, a hero, a mining-cycle panel, a Sensors page and an all-India coalfield map with 16 fields. Every screenshot in the deck predates all of it.
- **The demo script should lead with the refusals.** The 409 and the tamper detection are the moments that separate this from a CRUD app with a chatbot bolted on.
- **Gradle cannot run from an agent shell on this machine** — `Selector.open()` fails, so the launcher never reaches its daemon. JS-only changes now ship by swapping the Hermes bundle into the signed APK, which takes about three minutes instead of a full build. Native changes still need Android Studio.
- **Directives are polled every 12 seconds, not pushed.** Push needs FCM, a Google project and a network the demo does not have. Worth saying out loud before a judge asks, because the honest answer is good: it works in the same places the rest of the app works.
- **CLAUDE.md still describes the LiteRT-LM design.** Worth updating before anyone reads it as the spec.

Generated from the working system on 5 September 2026 — figures come from the live database, the running API and the handset, not from estimates. Print to PDF with Ctrl + P; the layout is set up for it.